FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

HIV/AIDS in Children

Tutorial | Week 2 · Diarrhoea, fluids, HIV & anaemia | Paediatrics | Dr Mawanda

Most Ugandan children with HIV were infected by their mothers. Without treatment about a third die before their first birthday, so every HIV-exposed infant needs prophylaxis, co-trimoxazole and DNA PCR at 6 weeks, 9 months and 6 weeks after weaning. Any positive child starts dolutegravir-based ART the same day, dosed by weight band (Uganda Clinical Guidelines 2023 and the 2022 Uganda consolidated HIV guidelines).

Must know before the ward round

1. Below 18 months, diagnose HIV by DNA PCR (virological test). Antibody tests detect the mother's IgG until about 18 months, so a positive rapid test in a baby proves exposure, not infection.
2. Uganda EID: 1st DNA PCR at 6 weeks (or first contact), 2nd at 9 months, 3rd 6 weeks after breastfeeding stops, then a rapid antibody test at 18 months. A positive PCR means start ART the same day and send a confirmatory sample at the same time.
3. Without ART about 30% of infected infants die by 1 year and 50% by 2 years (UCG 2023). Treat every child with HIV, whatever the CD4 count or WHO stage.
4. Preferred first line (Uganda 2022/UCG 2023): ABC + 3TC + DTG for children ≥4 weeks and ≥3 kg; DTG 50 mg film-coated tablet from 20 kg; TLD (TDF/3TC/DTG) from 30 kg.
5. Paediatric DTG (pDTG) 10 mg dispersible, once daily: 6–9.9 kg 1.5 tabs; 10–13.9 kg 2; 14–19.9 kg 2.5; 20–24.9 kg 3 (or one DTG 50 mg tablet). 3–5.9 kg: UCG table lists 1 tab, but the FDA label and WHO give 5 mg (½ tab); check the current MoH dosing chart.
6. ABC/3TC 120/60 mg dispersible, once daily: 3–5.9 kg 1 tab; 6–9.9 kg 1.5; 10–13.9 kg 2; 14–19.9 kg 2.5; 20–24.9 kg 3; ≥25 kg ABC/3TC 600/300 one tab.
7. On rifampicin-based TB treatment, give the weight-band DTG dose TWICE daily, and continue for 2 weeks after rifampicin stops.
8. Co-trimoxazole prophylaxis from 6 weeks for every HIV-exposed infant until infection is excluded; for life in HIV-positive children (UCG 2023). Once daily: <5 kg 120 mg; 5–14.9 kg 240 mg; 15–29.9 kg 480 mg; ≥30 kg 960 mg.
9. Viral load in children and adolescents (0–19 y): at 6 and 12 months after starting ART, then every 6 months. Uganda: ≤200 copies/mL suppressed; 201–999 low-level viraemia; ≥1000 non-suppressed → 3 intensive adherence counselling (IAC) sessions, repeat VL, resistance test, switch.
10. Presumptive severe HIV disease in an antibody-positive infant under 18 months whose PCR result is not yet back: oral thrush + severe pneumonia or severe sepsis, or any AIDS-defining illness (PCP, oesophageal candida, cryptococcal meningitis, severe wasting). Start ART and confirm by PCR.
11. PCP: infant aged 2–6 months, severe hypoxia with a fairly clear chest, little fever. Treat with high-dose co-trimoxazole (TMP 8 mg/kg/dose) three times a day for 21 days, plus prednisolone 1–2 mg/kg/day if hypoxic.
12. Disclosure is a process: partial disclosure ("you have germs in your blood that the medicine keeps asleep") from about 6–8 years, and full disclosure by about 10–12 years, before adolescence and sexual debut.

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1. Definition

Paediatric HIV is HIV infection in children under 15 years. About **90%** is **vertical** (mother-to-child): in utero, during birth or through breastfeeding. The rest comes from **sexual abuse**, adolescent sexual activity, unsafe injections or (rarely now) transfusion.

HIV runs **much faster in infants than in adults**. Without treatment about **30% die by age 1 and 50% by age 2** (UCG 2023).

Term	Meaning
HIV-exposed infant (HEI)	Infant born to a mother with HIV, until infection is excluded after breastfeeding ends (final rapid test at 18 months or later)
EID	Early infant diagnosis by virological testing (DNA PCR on DBS, or point-of-care)
Virological test	Detects the virus itself (DNA PCR, RNA viral load). The only valid test for diagnosis under 18 months
Serological test	Detects antibodies (rapid tests). Valid for diagnosis from **18 months**, once maternal antibody has gone and the child has stopped breastfeeding for at least 6 weeks
Presumptive severe HIV disease	Clinical diagnosis in an antibody-positive infant under 18 months when PCR is not yet available (criteria in section 5)
Advanced HIV disease (AHD)	WHO: **all children under 5 years** with HIV (at presentation or re-engagement), or CD4 <200 cells/µL (≥5 years), or WHO stage 3 or 4
Viral suppression (Uganda)	VL **≤200 copies/mL**. Low-level viraemia 201–999. Non-suppressed **≥1000**
Virological failure	Two consecutive VLs **>1000 copies/mL** 3–6 months apart, despite intensive adherence support, after at least 6 months on ART
Perinatally acquired HIV (PHIV)	Infection acquired from the mother; many such children are now adolescents

The whole programme rests on four things:

1. **Find them early.** Early infant diagnosis (**EID**) by **DNA PCR**, plus testing every sick, malnourished or TB-affected child whose status is unknown.
2. **Treat at once.** ART on the **same day** for every positive child (test and treat), with **DTG-based** regimens dosed by **weight band**.
3. **Prevent the killers. Co-trimoxazole**, TB preventive therapy, immunisation, nets, nutrition.
4. **Keep them suppressed.** Adherence, viral load monitoring, age-appropriate **disclosure**, and an **adolescent** service that keeps teenagers in care.

> **CLINICAL PEARL**
> The one thing never to forget: **every child admitted to a Ugandan ward needs a known HIV status.** If it isn't documented, test the child (and the mother). A severely malnourished, TB-infected or "pneumonia not responding" infant who turns out to be HIV-positive is the classic missed diagnosis.

2. Epidemiology

- Uganda has roughly **1.4 million** people living with HIV; children under 15 are a small but stubborn fraction, around **80,000–90,000** (UNAIDS estimates) Verify *current UNAIDS figure*. Paediatric ART coverage and viral suppression **lag behind adults**. Children are the group most often **unsuppressed**, and adolescents are the group most often **lost to follow-up**.
- **New infant infections** come mainly from mothers who **seroconvert in pregnancy or breastfeeding**, mothers who **drop out of care**, and **unsuppressed** mothers (see *HIV in Pregnancy*). Final MTCT (including breastfeeding) has fallen to the mid single digits.
- **Prevalence among sick children is far higher than in the community.** HIV is common among children admitted with **SAM, TB and recurrent pneumonia**, which is why UCG says to test every child with these.
- **Commonest killers** of untreated children: **PCP, severe bacterial pneumonia, sepsis, diarrhoea, TB and malnutrition**.
- **Services**: ART clinics at HC III and above, **mother–baby care points** (MBCP) for HEI up to 18 months, **POC EID** at many HC IVs and hospitals, otherwise DBS goes via the **hub system** to CPHL. Specialist paediatric HIV care is at **Baylor-Uganda (Mulago)**, IDI, regional Centres of Excellence and mission hospitals such as Mengo.
- **Adolescents**: "Young people's" clinics, **Operation Triple Zero (OTZ)** clubs (zero missed appointments, zero missed drugs, zero viral load), **Y+ peer supporters** and the **Ariel/Children's clubs**.

> **UGANDA CONTEXT**
> In practice the barriers on a Ugandan ward are: **pDTG stock-outs** (children left on LPV/r or NVP regimens), **orphanhood** and grandparents as caregivers, school boarding (doses missed, stigma), **non-disclosure** to the child, and **weight-band doses not updated** as the child grows. Always weigh the child and check the dose against the current band at every visit.

3. Aetiology and Risk Factors

Cause: human immunodeficiency virus (HIV-1 in almost all Ugandan children), a retrovirus that infects CD4 T cells.

Routes of infection in children

Route	Share / timing	Notes
In utero	About a quarter of vertical transmission	Infant PCR may already be positive at birth
Intrapartum	About half of non-breastfeeding transmission	Prolonged rupture of membranes and high maternal VL raise the risk
Breastfeeding	Risk continues for as long as the baby breastfeeds and the mother's VL is detectable	Why EID continues until 6 weeks after weaning
Sexual abuse; adolescent sexual activity	The main non-vertical routes	Ask in every child whose mother is HIV-negative
Unsafe injections, transfusion	Rare now	Screened blood

Without any intervention, total MTCT is **15–45%** in breastfeeding populations.

Risk factors for vertical transmission (the "high-risk infant")

- Mother **diagnosed late** (third trimester or while breastfeeding), or **seroconverted** in pregnancy or breastfeeding.
- Mother on ART **<4 weeks** before delivery, or **VL >1000** in the 4 weeks before delivery; mother who has **dropped out of care**.
- **Mixed feeding** in the first 6 months.
- **Missed or incomplete infant prophylaxis.**

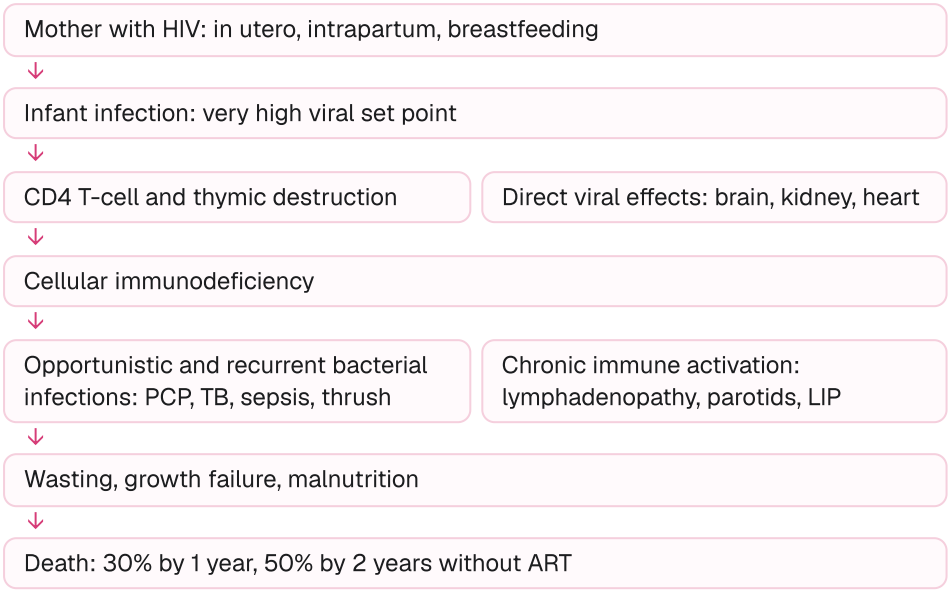
Risk factors for rapid progression and death in an infected child: infancy, no ART, no co-trimoxazole, malnutrition, TB, late diagnosis, poor adherence (caregiver change, orphanhood, non-disclosure).

4. Pathophysiology

Infection and injury. HIV infects CD4 T cells, macrophages and dendritic cells and destroys CD4 cells. In infants it seeds a **thymus that is still building the T-cell repertoire**, and the immune system has not yet met common pathogens.

Why infants progress fast. A very high viral set point (maternal antibody gives poor control), destruction of CD4 cells in a thymus still producing T cells, and an immature immune system. About **one in five** untreated infants are "rapid progressors" who die in the first year; most others progress over several years. A few (long-term non-progressors) present as adolescents with stunting, LIP, chronic lung disease or delayed puberty.

CD4 count varies with age. Infants normally have much higher counts than adults. Use **CD4 percentage** under 5 years (severe immunosuppression: <25% under 12 months, <20% at 12–35 months, <15% at 36–59 months) and absolute counts from 5 years (<200 cells/μL).



Feature links

- Loss of CD4 cells → failure of T-cell defence against *Pneumocystis* → PCP at 2–6 months: severe hypoxia with a fairly clear chest.
- Loss of T-cell control of *Candida* → oral thrush after 6 weeks; oesophageal candida → painful swallowing, refusal to feed.
- Poor cell-mediated immunity → TB (pulmonary, lymph node, disseminated, meningitis).
- B-cell dysregulation with poor specific antibody → recurrent bacterial pneumonia, sepsis, otitis, meningitis.
- Chronic immune activation and lymphoid hyperplasia → persistent generalised lymphadenopathy, hepatosplenomegaly, bilateral painless parotid enlargement, LIP (with clubbing).

- Recurrent infection + diarrhoea + raised metabolic demand + poor intake → faltering growth, stunting, SAM, delayed puberty.
- Virus in the developing brain → HIV encephalopathy: lost or delayed milestones, microcephaly, spastic diplegia.
- Chronic inflammation and marrow suppression → anaemia, neutropenia, thrombocytopenia.
- Virus and immune complexes in the kidney and heart → HIV nephropathy (oedema, proteinuria), cardiomyopathy.
- Loss of immune surveillance of oncogenic viruses (HHV-8, EBV) → Kaposi sarcoma, lymphoma.
- Rapid immune recovery on ART → inflammatory response to hidden pathogens → IRIS 2–12 weeks after starting ART.

Why DTG? Dolutegravir is an **integrase inhibitor** with a **high genetic barrier to resistance**, fast viral load reduction, good tolerability and a palatable once-daily dispersible tablet. Many Ugandan infants were exposed to **NVP** prophylaxis or to mothers with NNRTI resistance, so NNRTI-based regimens (NVP, EFV) fail more often. **LPV/r** works but tastes terrible and the syrup needs refrigeration.

Why co-trimoxazole? It prevents **PCP** (peak age 2–6 months, often the first sign of HIV), bacterial pneumonia, sepsis, toxoplasmosis, isosporiasis and **malaria**. It cuts mortality in HIV-positive children even on ART.

5. Classification / Staging

WHO clinical staging for children (2007, still in use)

Stage	Typical conditions (all "unexplained", i.e. no other cause)
1	Asymptomatic; persistent generalised lymphadenopathy
2	Persistent hepatosplenomegaly; papular pruritic eruptions; extensive warts or molluscum; fungal nail infection; recurrent oral ulcers; lineal gingival erythema; **parotid enlargement**; herpes zoster; recurrent or chronic URTI (otitis, sinusitis, tonsillitis)
3	**Moderate malnutrition** not responding to treatment; persistent diarrhoea ≥14 days; persistent fever >1 month; **persistent oral candidiasis (after 6 weeks of age)**; oral hairy leukoplakia; acute necrotising gingivitis; **lymph-node TB; pulmonary TB**; severe recurrent bacterial pneumonia; **symptomatic lymphoid interstitial pneumonitis (LIP)**; chronic HIV lung disease (bronchiectasis); anaemia <8 g/dL, neutropenia <0.5 × 10^9/L, platelets <50 × 10^9/L
4	**Severe wasting, stunting or SAM** not responding to treatment; **PCP**; recurrent severe bacterial infections (not pneumonia: empyema, pyomyositis, bone/joint infection, meningitis); chronic HSV (>1 month); **oesophageal candidiasis**; **extrapulmonary TB**; **Kaposi sarcoma**; CMV (onset after 1 month); CNS toxoplasmosis (after 1 month); **HIV encephalopathy**; **cryptococcal meningitis**; disseminated NTM; chronic cryptosporidiosis or isosporiasis; lymphoma; HIV nephropathy or cardiomyopathy; acquired recto-vaginal fistula

EXAM FAVOURITE

Oral thrush is **stage 3 only after 6 weeks of age** (thrush in a newborn is common and normal). Thrush **in the oesophagus** (pain on swallowing, refusing feeds) is **stage 4**.

Immunological classification (severe immunosuppression)

Age	Severe immunosuppression
<12 months	CD4 **<25%**
12–35 months	CD4 **<20%**

Age	Severe immunosuppression
36–59 months	CD4 **<15%**
≥5 years	CD4 **<200 cells/μL**

Advanced HIV disease: all children under 5 years, or CD4 <200 cells/μL (≥5 years), or WHO stage 3 or 4.

Presumptive severe HIV disease (antibody-positive infant under 18 months, PCR result not yet back): oral thrush + severe pneumonia or severe sepsis, **or** any AIDS-defining illness (PCP, oesophageal candida, cryptococcal meningitis, severe wasting). Start ART and confirm by PCR.

Viral load categories (Uganda): ≤200 copies/mL suppressed; 201–999 low-level viraemia; ≥1000 non-suppressed.

6. Clinical Features

History

Ask	Why it matters
Mother's HIV status, when diagnosed, ART, last viral load	Defines exposure and risk; a mother diagnosed late or unsuppressed = high-risk infant
Infant prophylaxis (NVP ± AZT) and how long	Missed prophylaxis raises risk; NVP exposure affects regimen choice
Feeding: exclusive breastfeeding? mixed? still breastfeeding? when stopped?	Ongoing exposure; mixed feeding in the first 6 months raises transmission; decides timing of the next PCR
Previous PCR / rapid test results and dates (check the HEI card and register)	Where the child sits on the EID algorithm
Co-trimoxazole started at 6 weeks?	Missed CTX is common; PCP risk
Symptoms: fever, cough, breathing difficulty, diarrhoea, poor feeding, **painful swallowing**, headache, convulsions, rash	Staging and OIs (painful swallowing = oesophageal candida, stage 4)
Infections: recurrent pneumonia, diarrhoea ≥14 days, ear discharge, thrush, skin rash, fever >1 month	WHO staging and OI risk
Growth: weight trajectory on the child health card	Faltering growth is often the earliest sign
TB contact or symptoms (cough, fever, poor weight gain, night sweats)	TB is the commonest OI; changes ART dosing and timing
Development: milestones lost or delayed	**HIV encephalopathy** (stage 4)
If on ART: regimen, dose, who gives it, missed doses, vomiting, how stored, school/boarding, who knows, VL history	Adherence is the main reason for failure; failure versus new illness
Disclosure: does the child know? does anyone else know?	Plan the disclosure process; adolescent risk
Adolescents (HEADSSS): home, education, activities, drugs, sexuality, suicide/depression, safety	Sexual risk, mental health, abuse, transition
Immunisation status and vitamin A	BCG and live vaccines in symptomatic HIV; catch-up
Siblings and other family members tested?	Index-client testing ("Know Your Child's Status")

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), adapted for children: observe first, examine the chest while the child is quiet, and leave the mouth and ears to last. In a sick child the ETAT/ABCDE assessment (section 9) comes first.

Introduction and preparation

- Introduce yourself; confirm the child's name, age and the caregiver's relationship (often a grandparent).
- Explain the examination; consent from the caregiver and assent from the child. Be discreet about HIV status in front of others.
- Hand hygiene.
- Position and exposure: on the caregiver's lap; undress fully but keep the child warm; adolescents examined with a chaperone and privacy.

General inspection from the end of the bed

- Well or unwell; **danger signs** (lethargy, unable to drink, convulsions, severe respiratory distress).
- Colour: pallor, cyanosis; hydration.
- **Nutrition**: visible wasting, oedema (SAM or HIV nephropathy), sparse hair.
- Behaviour and development (head control, sitting, interaction): encephalopathy.
- **Vitals**: RR (fast breathing ≥60 under 2 months, ≥50 at 2–11 months, ≥40 at 1–5 years), HR, **SpO_2**, temperature, capillary refill.
- **Growth**: weight, length/height, **MUAC** and **head circumference**, plotted as **Z-scores**. Stunting is very common; microcephaly suggests encephalopathy.

Hands, face and neck

Sign	What it means
Clubbing	LIP, bronchiectasis
Fungal nail infection (fingers and toes)	Stage 2
Pallor of palms and conjunctivae	Anaemia (HIV, malaria, AZT)
Bilateral painless parotid enlargement	Think **LIP** (stage 2)
Mouth (examine last in a young child): **thrush**, angular cheilitis, gingivitis, lineal gingival erythema, ulcers, **hairy leukoplakia** (adolescents), Kaposi lesions on the palate	Thrush after 6 weeks = stage 3; Kaposi = stage 4
Persistent generalised lymphadenopathy (cervical, axillary, inguinal)	Stage 1; large, hard, progressive nodes → TB or lymphoma
Neck stiffness, bulging fontanelle	Cryptococcal or TB meningitis

Skin: papular pruritic eruption, seborrhoeic dermatitis, extensive molluscum or warts, herpes zoster or its scars, scabies, Kaposi nodules, **drug rash** (abacavir hypersensitivity, NVP, co-trimoxazole), BCG scar.

Systems

- **Respiratory**: inspection (RR, chest indrawing, nasal flaring), palpation, percussion (dull: consolidation or effusion), auscultation (crackles, bronchial breathing, wheeze). A **hypoxic infant with a relatively clear chest** = PCP until proven otherwise.
- **Cardiovascular**: gallop, murmur, hepatomegaly (cardiomyopathy).
- **Abdomen**: hepatosplenomegaly, ascites, oedema.
- **Neurological and development**: milestones, tone (spastic diplegia in encephalopathy), reflexes, neck stiffness, fontanelle.
- **Adolescents**: Tanner stage (puberty is often delayed), signs of STI, pregnancy.

- **Ears last**: ear discharge (chronic suppurative otitis media).

Complete the examination: plot growth, check the HEI or ART card, thank the caregiver and child, wash hands, and summarise (model presentation in section 8).

Document negatives: "no thrush, no lymphadenopathy, no hepatosplenomegaly, no signs of meningism".

Red flags

RED FLAG

- **Danger signs (IMCI/ETAT)**: unable to drink or breastfeed, vomits everything, convulsions, lethargy or unconsciousness, plus SpO_2 <90%, severe respiratory distress, shock.
- **Severe pneumonia in an infant aged 2–6 months** who is HIV-exposed: add **high-dose co-trimoxazole for PCP** to the antibiotics.
- **Headache, vomiting, confusion or neck stiffness** in an older child or adolescent on ART: **cryptococcal meningitis** or TB meningitis.
- **Rash with fever, abdominal pain or breathing difficulty within 6 weeks of starting abacavir**: hypersensitivity. Stop it and **never re-challenge**.
- **Clinical deterioration 2–12 weeks after starting ART**: IRIS (often unmasked TB). Do not stop ART.
- **SAM** (MUAC <11.5 cm, W/H <–3 Z or oedema) in an HIV-positive child: high mortality; treat the SAM, start ART within 2 weeks once stable.
- **Teenager who has stopped coming**: high risk of AHD, depression, pregnancy, death.

Thinking like a clinician

- **"Positive rapid test at 9 months" does not mean infected.** It means **exposed**. Do a DNA PCR. Conversely, a **negative** rapid test from 9 months in a child who is **not breastfeeding** makes infection very unlikely.
- **"PCR negative at 6 weeks, so the child is negative."** Only if the child **never breastfed**. If still breastfeeding, the child stays exposed: CTX, next PCR at 9 months, and 6 weeks after weaning.
- **If an infant fails to thrive, has persistent thrush after 6 weeks, or recurrent pneumonia, test for HIV even if a previous PCR was negative.** UCG says do a PCR any time an exposed child becomes symptomatic.
- **Malnutrition + HIV + TB travel together.** Screen every malnourished child for both.
- **On ART but not suppressed? Think adherence first** (taste, vomiting, caregiver change, school, stigma, the child secretly spitting out tablets), **then dosing** (weight band out of date), **then interactions** (rifampicin, carbamazepine, antacids), **then resistance**.
- **A child "on ART for years" with a new stage 3/4 illness** after 6 months on effective ART = **clinical failure** (or IRIS if just started). Check a viral load.
- **Do not delay ART for CD4 or "readiness" in an infant.** Same-day ART is the default; the exceptions are **TB meningitis and cryptococcal meningitis**, where ART is deferred 4–6 weeks.
- **If the infant is 2–6 months old, breathless and hypoxic but the chest sounds nearly clear, think PCP**, and give high-dose co-trimoxazole with the antibiotics.
- **Common student mistakes**: reading a positive rapid test under 18 months as infection; forgetting that breastfeeding keeps the baby exposed; swapping pDTG for DTG 50 mg mg-for-mg; not re-weighing and re-banding the dose; stopping co-trimoxazole once ART starts.

7. Differential Diagnoses

When an HIV-exposed or HIV-positive child presents with the "HIV look" (wasting, recurrent infections, lymphadenopathy), think about these:

Differential	Distinguishing clinical feature	Distinguishing investigation
TB (pulmonary/disseminated)	Contact history, fever >2 weeks, weight loss, hilar nodes, no response to antibiotics	GeneXpert (sputum/gastric/stool), CXR, TB-LAM
SAM without HIV	Dietary history, oedema, responds to F-75/F-100/RUTF	HIV test is negative
PCP vs bacterial pneumonia	PCP: 2–6 months, little fever, severe hypoxia, clear-ish chest, no response to 48 h antibiotics	SpO_2, CXR (ground glass), LDH
LIP vs miliary TB	LIP: older child (>2 y), clubbing, parotids, slow course, well despite CXR	CXR pattern, Xpert, trial of antibiotics
Congenital infection (CMV, syphilis, toxoplasmosis)	Hepatosplenomegaly, rash, jaundice from birth	Maternal RPR, infant serology
Primary immunodeficiency	Family history, recurrent infections, HIV PCR negative	Immunoglobulins (referral)
Malignancy (lymphoma, leukaemia, Kaposi)	Lymph nodes that are big, hard and progressive; cytopenias	FBC/film, biopsy
Chronic suppurative lung disease after measles/pertussis	History of the infection, clubbing	CXR, HIV test

8. Clinical Workup

Diagnosis depends on age: **virological (DNA PCR) under 18 months, serological (rapid tests) from 18 months**. Treat emergencies first; never delay ART for tests other than those that change its timing (TB meningitis, cryptococcal meningitis).

Bedside

Facility levels in these tables are compiled from the source notes Verify *against the MoH test menu by level*.

Test	Why	Expected / abnormal finding	Facility level
HIV rapid test (child ≥18 months; or mother; or infant to establish exposure)	Diagnosis ≥18 months; exposure under 18 months	Positive ≥18 months (and not breastfed for 6 weeks) = infected; under 18 months = exposed only	HC II and above
POC EID (DNA PCR) where available	Same-day infant diagnosis	Positive = infected (send a confirmatory sample)	Many HC IVs and hospitals
RBS	Hypoglycaemia in the sick or malnourished child	<3.0 mmol/L (UCG); WHO <2.5 (<3.0 in SAM): 10% dextrose 5 mL/kg	All levels with a glucometer
Hb (HemoCue)	Anaemia (HIV, malaria, AZT)	<8 g/dL = stage 3	HC III and above
Malaria RDT	Fever; CTX reduces but does not abolish malaria	Positive: treat malaria	All levels

Test	Why	Expected / abnormal finding	Facility level
SpO_2	PCP, pneumonia	<90%: oxygen; very low with clear chest = PCP	HC IV and above
Urine TB-LAM	Advanced disease or seriously ill (WHO); UCG lists CD4 <200	Positive: start TB treatment	HC III/IV and above
Serum/plasma CrAg (adolescents with CD4 <200)	Pre-emptive fluconazole before cryptococcal meningitis develops	Positive: LP, then pre-emptive or full treatment	HC IV and above
Weight, height, MUAC, Z-scores	Dose band; malnutrition; staging	MUAC <11.5 cm = SAM	All levels

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Availability
DNA PCR (DBS)	HEI at 6 weeks, 9 months, 6 weeks after weaning; any symptomatic HEI	Positive = infected (send a confirmatory sample)	Start ART **today**; do not wait for the confirmatory result	DBS from HC II via hub to CPHL (days–weeks)
Rapid antibody tests (Determine → Stat-Pak → SD Bioline tie-breaker per national algorithm)	Diagnosis ≥18 months; final test at 18 months	Positive = infected (≥18 months and not breastfed for 6 weeks)	Start ART	HC II and above
CD4 count / %	Baseline; suspected treatment failure	Low = AHD	AHD package: CrAg, TB-LAM, closer follow-up	HC IV and above (POC CD4)
Viral load (DBS or plasma)	6 and 12 months on ART, then every 6 months (0–19 y)	≤200 suppressed; 201–999 LLV; ≥1000 non-suppressed	IAC, repeat VL, HIVDR test, switch	DBS from HC II via hub; results take weeks
FBC	Baseline; on AZT; anaemia	Anaemia, neutropenia (AZT, CTX), thrombocytopenia	Change AZT; investigate; staging	HC IV and above
Creatinine / urinalysis	Adolescents on TDF; oedema	Proteinuria, raised creatinine	TDF contraindicated if CrCl <60 mL/min (UCG)	Hospital
HBsAg	Adolescents before TDF	Positive: HBV co-infection	Regimen choice (keep TDF + 3TC)	HC IV and above
GeneXpert (sputum, gastric aspirate, stool)	TB symptoms	MTB detected ± RIF resistance	TB treatment; DTG twice daily	HC IV and hospitals; samples referred from lower levels

Imaging

Test	When	Expected finding	How it changes management	Availability
Chest X-ray	Pneumonia not responding, suspected TB, LIP	**PCP**: bilateral ground glass; **LIP**: reticulonodular + hilar nodes; **TB**: hilar nodes, consolidation, miliary	Directs treatment	District/regional hospitals

Specialised / referral-level

Test	When	Expected finding	How it changes management	Availability
HIV drug resistance (genotype)	Two non-suppressed VLs on a DTG or PI regimen; mother failing at baby's diagnosis	Mutations	Choose 2nd/3rd-line regimen	CPHL via hub
Lumbar puncture with CrAg and India ink, TB Xpert	Headache, meningism, confusion	CrAg positive; high opening pressure	Cryptococcal or TB meningitis protocol; defer ART	Hospital
LDH	Suspected PCP	Raised	Supports PCP	RRH, NRH
Immunoglobulins	HIV-negative child with recurrent infections	Low	Primary immunodeficiency referral	NRH
Biopsy (node, skin)	Hard progressive nodes, skin nodules	TB, lymphoma, Kaposi sarcoma	Specific treatment	RRH, NRH

Investigation summary

Test	What you are looking for	Why
DNA PCR / rapid tests (age-appropriate)	Infection	Diagnosis, start ART
CD4 (% under 5 years)	Degree of immunosuppression	AHD, OI risk
Viral load	Treatment response	Detects failure early
FBC	Anaemia, neutropenia, thrombocytopenia	Staging, AZT/CTX toxicity
GeneXpert, CXR, TB-LAM	TB	TB treatment and ART dosing
CrAg (adolescents, CD4 <200), LP if symptoms	Cryptococcus	Pre-emptive fluconazole or treatment
HBsAg (adolescents before TDF), creatinine	HBV, kidney function	Regimen choice
Growth (Z-scores), development	Malnutrition, encephalopathy	Nutrition support, staging

Putting it together

Model one-liner

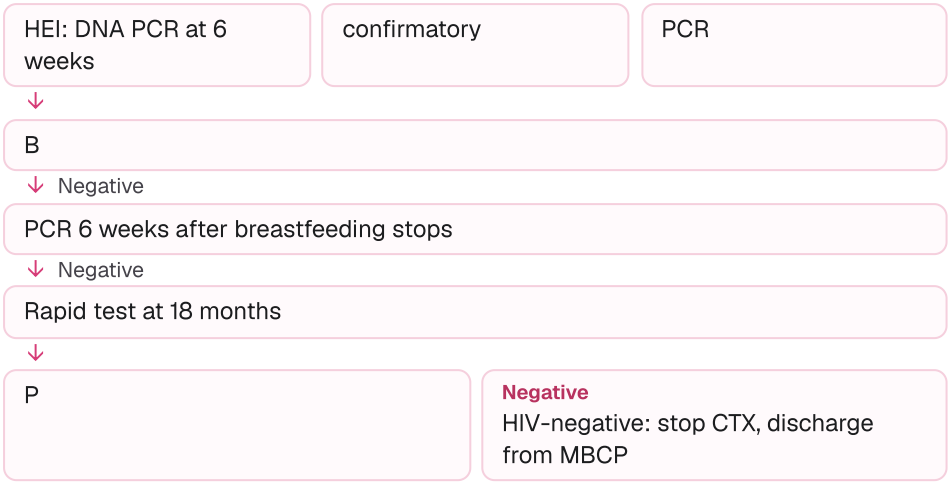
> "Baby A, an 8-month-old HIV-exposed boy, weight 5.9 kg (WAZ –2.6, MUAC 11.8 cm), mixed-fed, whose mother was diagnosed in the third trimester and is not suppressed, presents with 3 days of fast breathing and oral thrush; he has severe pneumonia (SpO_2 86%) and moderate acute malnutrition. His 6-week PCR was never collected."

Problem list

1. Severe pneumonia with hypoxaemia, likely bacterial ± PCP.
2. Presumptive severe HIV disease.
3. MAM/faltering growth.
4. Missed EID and CTX.
5. Unsuppressed mother.

Worked interpretation: the EID algorithm

- 6-week PCR **negative**, baby **exclusively breastfed and still breastfeeding** → still exposed. Continue CTX, NVP as prescribed, **PCR at 9 months**.
- 9-month PCR **negative**, still breastfeeding → **PCR 6 weeks after complete cessation of breastfeeding**, then **rapid test at 18 months**.
- Rapid test at 18 months **negative** and not breastfed for 6 weeks → **HIV-negative**: stop CTX and discharge from the MBCP.
- Any **positive PCR** → "HIV-positive": start ART the same day, send a **confirmatory** DBS (and an HIVDR sample if the mother is failing).



Model case presentation (ward round)

> "Akello is a 2-year-4-month-old girl from Mukono, brought by her grandmother, admitted with 2 weeks of cough and fever and 3 days of fast breathing. She weighs 8.6 kg (WAZ −2.9), height 80 cm (HAZ −2.8), WHZ −1.9, MUAC 12.2 cm. Her mother died last year; Akello was diagnosed HIV-positive at 14 months and is on ABC/3TC 120/60 two tablets plus pDTG 10 mg two tablets daily, but the grandmother says doses are 'sometimes missed' since she started at nursery. Her last VL 4 months ago was 3,200 copies/mL. She is fully immunised for age except MR2 (due at 18 months, missed) and was breastfed until 9 months. She has a TB contact (uncle with cough). On examination: RR 52, SpO_2 92%, no danger signs, oral thrush, generalised lymphadenopathy, bilateral parotid enlargement and hepatomegaly 3 cm. My working diagnosis is pneumonia with suspected TB, in a child with WHO stage 3 HIV who is not suppressed, likely from poor adherence, with moderate wasting and stunting. Plan: Xpert on gastric aspirate and stool, CXR, urine TB-LAM, CD4 and FBC, amoxicillin, nutrition referral, and intensive adherence counselling with the grandmother; her ART dose band is correct for 8.6 kg."

9. Management

Immediate / emergency

First 5 minutes

1. **ETAT triage**: emergency signs (Airway/Breathing, Circulation, Coma, Convulsions, severe Dehydration). Treat these first, whatever the HIV status.
2. Vitals with age-appropriate norms: RR (fast breathing ≥60 under 2 months, ≥50 at 2–11 months, ≥40 at 1–5 years), HR, SpO_2, temperature, **RBS**, capillary refill.
3. **Weigh** the child now: every drug in this topic is dosed by weight.

ABCDE

- **A/B**: oxygen if SpO_2 <90%. If an HIV-exposed infant aged 2–6 months has **severe pneumonia**: start ampicillin + gentamicin **and** high-dose co-trimoxazole for PCP at once (plus prednisolone if hypoxic).
- **C**: treat shock per ETAT (care in SAM: follow the SAM fluid protocol).
- **D**: treat convulsions; treat hypoglycaemia (**10% dextrose 5 mL/kg IV**). Headache, confusion or neck stiffness: LP for cryptococcal and TB meningitis.
- **E**: temperature; rash (drug reaction: abacavir, NVP, co-trimoxazole); malaria RDT.

Definitive

A. Care of the HIV-exposed infant (mother–baby care point, up to 18 months)

1. **ARV prophylaxis**, started within 6 hours of birth (see *HIV in Pregnancy*):
   - **Low risk**: **NVP** once daily for **6 weeks**.
   - **High risk** (mother on ART <4 weeks before delivery, VL >1000 in the 4 weeks before delivery, or diagnosed in the 3rd trimester or while breastfeeding): **AZT + NVP for 6 weeks, then NVP alone for 6 more weeks** (12 weeks in total), per the 2022 consolidated guidelines.
   - **Baby first seen after 6 weeks** (UCG 2023): take a PCR, give a **first-line paediatric ART regimen for 6 weeks**; if the PCR is negative, follow with NVP for 6 weeks; if positive, continue ART.
2. **DNA PCR** at 6 weeks, 9 months and 6 weeks after weaning; rapid test at 18 months (algorithm in section 8).
3. **Co-trimoxazole from 6 weeks** until infection is excluded and breastfeeding has stopped.
4. **Immunise** per UNEPI (see *Immunisation*). Give BCG at birth; do **not** give BCG later to a child with **symptomatic HIV**.
5. **Feeding**: exclusive breastfeeding for 6 months, then complementary foods, with breastfeeding continued to **12 months** or beyond while the mother is suppressed on ART. **Never mix-feed** under 6 months.
6. **Visits** synchronised with immunisation: 6, 10, 14 weeks, then 5, 6, 9, 12, 15 and 18 months (UCG 2023). Weigh, plot, screen for TB and illness every visit.

> **UGANDA CONTEXT**
> High-risk infant prophylaxis: the **2022 consolidated guidelines** give **AZT + NVP for 6 weeks, then NVP for 6 weeks**; **UCG 2023** lists **NVP alone for 12 weeks**. Follow your unit's protocol.

B. Diagnosis made: starting ART

Principles (UCG 2023): test and treat everyone; **same-day ART** after counselling the caregiver (if not ready, within **7 days** for children); screen for TB and cryptococcal disease first, because ART is **deferred** in TB meningitis and cryptococcal meningitis.

Before starting:

1. Confirm the diagnosis (but **do not wait** for the confirmatory PCR to start).
2. Weigh, measure, stage (WHO), screen for TB (symptoms ± Xpert), check CD4 (to identify AHD), FBC.
3. Identify the **treatment supporter** (usually the mother or a grandparent) and teach how to disperse and give the tablets.
4. Plan disclosure (age-appropriate).

Recommended first-line regimens (Uganda 2022 consolidated guidelines / UCG 2023)

Group	Preferred	Alternatives
Neonates (<4 weeks)	**AZT + 3TC + NVP** (or RAL-based where available) Verify *current neonatal option*	Switch to ABC + 3TC + DTG once ≥4 weeks and ≥3 kg
Children ≥4 weeks and <20 kg	**ABC + 3TC + DTG** (pDTG 10 mg dispersible, or ABC/3TC/DTG 60/30/5 mg dispersible FDC)	If DTG not tolerated or not available: **ABC + 3TC + LPV/r** granules/pellets; if LPV/r not tolerated: ABC + 3TC + EFV (only >3 years and >10 kg); if ABC contraindicated: AZT + 3TC + DTG (or LPV/r)
Children 20 to <30 kg	**ABC + 3TC + DTG** (DTG 50 mg film-coated tablet)	If DTG contraindicated: ABC + 3TC + LPV/r tablets; if ABC contraindicated: TAF/FTC/DTG (>6 years and >25 kg) or AZT + 3TC + DTG
Adolescents ≥30 kg	**TDF + 3TC + DTG (TLD)** one tablet daily	TAF/FTC/DTG; ABC/3TC + DTG if TDF contraindicated (renal disease, CrCl <60)

DRUG DOSING

Abacavir hypersensitivity (fever, rash, GI or respiratory symptoms, usually in the first 6 weeks): stop ABC and **never restart it**; the reaction on re-challenge can be fatal. HLA-B*5701 is rare in African children, but the reaction can still occur.

Weight-band dosing (UCG 2023 dosing tables; once daily unless stated)

Formulation	3.0–5.9 kg	6.0–9.9 kg	10.0–13.9 kg	14.0–19.9 kg	20.0–24.9 kg	25.0–34.9 kg
ABC/3TC 120/60 mg dispersible, once daily	1	1.5	2	2.5	3	Use 600/300
ABC/3TC 600/300 mg, once daily	–	–	–	–	–	**1**
pDTG 10 mg dispersible, once daily	1 per UCG table (FDA/WHO: **5 mg = ½ tab**) Verify *current MoH dosing chart*	**1.5** (15 mg)	**2** (20 mg)	**2.5** (25 mg)	**3** (30 mg)	Use 50 mg
DTG 50 mg film-coated, once daily	–	–	–	–	**1**	**1**
ABC/3TC/DTG 60/30/5 mg dispersible, once daily	2	3	4	5	6	–
AZT/3TC 60/30 mg, twice daily	1 + 1	1.5 + 1.5	2 + 2	2.5 + 2.5	3 + 3	AZT/3TC 300/150 1 + 1
LPV/r 40/10 mg pellets/granules, **twice daily**	2 + 2	3 + 3	4 + 4	5 + 5	6 + 6	–
LPV/r 100/25 mg tablets, **twice daily**	–	–	2 + 1	2 + 2	2 + 2	Use 200/50: 2 + 1
EFV 200 mg, at night (>3 y, >10 kg)	–	–	1	1.5	1.5	2

How the doses relate to mg/kg: DTG works out at roughly **1.5–2.5 mg/kg/day** (max **50 mg** once daily, or 50 mg twice daily with rifampicin); ABC about **16 mg/kg/day** (max **600 mg**/day); 3TC about **8 mg/kg/day** (max **300 mg**/day); AZT about **180–240 mg/m² twice daily** (max **300 mg** per dose). Use the **band**, not a calculation, on the ward.

DRUG DOSING

pDTG 10 mg dispersible and DTG 50 mg film-coated tablets are not bioequivalent. 30 mg of pDTG equals 50 mg of the film-coated tablet. Never substitute mg for mg. Disperse pDTG in 5 mL of clean water (up to 3 tablets) or 10 mL (4–6 tablets) and give at once.

DRUG DOSING

DTG interactions: give **twice daily** with **rifampicin** (same weight-band dose, morning and evening) and continue for 2 weeks after rifampicin ends. Give DTG **2 hours before or 6 hours after** antacids, iron, calcium or zinc, unless taken together with food. Avoid with carbamazepine, phenytoin and phenobarbital (UCG lists these as contraindications: use another antiepileptic such as levetiracetam or valproate).

C. Co-trimoxazole prophylaxis (CPT)

Weight (UCG 2023 / WHO bands)	Daily dose (SMX/TMP)	Practical form
<5 kg (usually 6 weeks–6 months)	**120 mg** (TMP about 4 mg/kg)	2.5 mL of 240 mg/5 mL syrup or ½ × 240 mg dispersible
5–14.9 kg	**240 mg**	5 mL syrup or ½ × 480 mg tablet
15–29.9 kg	**480 mg**	1 × 480 mg tablet
≥30 kg	**960 mg**	1 × 960 mg or 2 × 480 mg

Oral, **once daily**; for HEI, from **6 weeks until HIV is excluded** after breastfeeding ends; for HIV-positive children, **for life** (UCG 2023). If CTX is contraindicated (severe sulfa reaction), give **dapsone 2 mg/kg once daily (max 100 mg)**. CTX reduces malaria but does not abolish it: a febrile child on CTX still needs a malaria test.

D. TB preventive therapy and TB co-treatment

- **Exclude TB disease** first (symptom screen ± Xpert/CXR).
- **Isoniazid 10 mg/kg (range 7–15) once daily, max 300 mg, plus pyridoxine, for 6 months** for HEI or HIV-positive children with TB contact (UCG 2023). Older children and adolescents may receive rifapentine-based short courses (3HP) per the current national TPT guidance Verify *eligibility and age limits*.
- **Child with TB and HIV**: start TB treatment first, then **ART within 2 weeks** (as soon as TB drugs are tolerated); **TB meningitis: wait 4–8 weeks**. On rifampicin, **DTG twice daily**; for LPV/r, add extra ritonavir ("super-boosting") to a 1:1 ratio; NVP must not be given with rifampicin. (Under the WHO 2022 TB guidelines, children with non-severe TB can receive the 4-month regimen; see *Childhood TB*.)

E. Opportunistic infections

- **PCP**: high-dose co-trimoxazole (IV preferred) three times daily for 21 days + prednisolone if SpO_2 <90% or severe distress; then back to prophylaxis for life. Pentamidine if severe sulfa reaction.
- **Oral thrush**: nystatin; miconazole gel if it fails. **Oesophageal candidiasis**: fluconazole.
- **Cryptococcal meningitis**: WHO 2022 induction (single high-dose liposomal amphotericin + flucytosine + fluconazole), then consolidation and maintenance fluconazole; therapeutic LPs; defer ART 4–6 weeks.
- **CrAg positive without meningitis** (adolescents): pre-emptive fluconazole after an LP if possible.
- **Symptomatic, hypoxic LIP**: prednisolone, after ruling out TB.

All doses are in the drug table below.

UGANDA CONTEXT

Cryptococcal meningitis: the WHO 2022 single high-dose liposomal amphotericin regimen was proven in the AMBITION-cm trial, largely run in Uganda. Where L-AmB is not available, the older regimen is used: **amphotericin B deoxycholate 1 mg/kg/day + flucytosine for 1 week, then fluconazole 12 mg/kg/day for 1 week**, followed by consolidation. Do **therapeutic LPs** for raised pressure. **Defer ART 4–6 weeks** after starting antifungals.

Drug table

ARV doses are by weight band (table above); this table covers prophylaxis and OI treatment.

Drug	Dose (per kg + max for children; adult)	Route	Frequency	Duration	Key cautions
Nevirapine syrup 10 mg/mL (prophylaxis, UCG 2023)	Birth–6 weeks: **10 mg (1 mL)** if 2.0–2.49 kg; **15 mg (1.5 mL)** if ≥2.5 kg. 6–12 weeks: **20 mg (2 mL)**	Oral	Once daily	6 weeks (12 if high-risk)	Start within 6 hours of birth; never with rifampicin
Zidovudine syrup 10 mg/mL (high-risk prophylaxis)	≥2.5 kg: **15 mg (1.5 mL)** (about 4 mg/kg); 2.0–2.49 kg: 10 mg (1 mL)	Oral	**Twice** daily	6 weeks	Anaemia, neutropenia
Co-trimoxazole (prophylaxis)	<5 kg **120 mg**; 5–14.9 kg **240 mg**; 15–29.9 kg **480 mg**; ≥30 kg and adult **960 mg**	Oral	Once daily	HEI: from 6 weeks until HIV excluded; HIV-positive: for life	Rash (Stevens–Johnson), neutropenia; still test for malaria
Dapsone (if CTX contraindicated)	**2 mg/kg**, max 100 mg	Oral	Once daily	As for CTX	Haemolysis in G6PD deficiency
Co-trimoxazole, high dose (PCP)	**TMP 8 mg/kg + SMX 40 mg/kg** per dose (WHO Pocket Book 2013)	IV (preferred) or oral	**Three times daily**	**21 days**	Then back to prophylaxis for life
Prednisolone (PCP)	**1–2 mg/kg/day**, max 40–60 mg/day	Oral	Once daily	1 week (then taper)	If SpO_2 <90% or severe distress
Pentamidine (PCP)	4 mg/kg	IV infusion	Once daily	21 days	For severe sulfa reaction
Nystatin 100,000 U/mL (oral thrush)	1–2 mL	Oral	Four times daily	7 days	Miconazole gel if it fails
Fluconazole (oesophageal candidiasis)	**3–6 mg/kg**, max 400 mg	Oral	Once daily	7 days (Pocket Book), 14–21 days in severe disease	Hepatotoxicity; interactions
Liposomal amphotericin B (cryptococcal meningitis, WHO 2022)	**10 mg/kg single dose**	IV	Once	Day 1	Where available (regional/national referral)
Flucytosine (cryptococcal meningitis)	**100 mg/kg/day** in 4 divided doses	Oral	6-hourly	14 days	Marrow suppression

Drug	Dose (per kg + max for children; adult)	Route	Frequency	Duration	Key cautions
Fluconazole (cryptococcal induction)	**12 mg/kg/day**, max 800 mg	Oral	Once daily	14 days	With L-AmB + flucytosine
Fluconazole (cryptococcal consolidation)	6–12 mg/kg/day, max 800 mg	Oral	Once daily	8 weeks	
Fluconazole (cryptococcal maintenance)	6 mg/kg/day, max 200 mg	Oral	Once daily	Until immune recovery	
Fluconazole (CrAg positive, no meningitis; adolescents)	12 mg/kg/day for 2 weeks, then 6 mg/kg/day for 8 weeks, then maintenance; max 800 mg/day, then 400 mg/day	Oral	Once daily	10 weeks, then maintenance	Do an LP first if possible
Prednisolone (LIP, symptomatic, hypoxic)	**1–2 mg/kg/day**	Oral	Once daily	2 weeks, taper over 2–4 weeks	Rule out TB first
Isoniazid (TPT)	**10 mg/kg** (range 7–15), max 300 mg	Oral	Once daily	6 months	Add pyridoxine; exclude TB disease first
10% dextrose (hypoglycaemia)	5 mL/kg	IV	Once, recheck glucose	As needed	Then feed

Referral criteria

Verify *level split compiled from the source notes and services described above, not an MoH list*

Level	What to do	Refer up when
HC II / HC III	HIV testing, HEI care at the MBCP, DBS collection, CTX, ART initiation and refills (ART clinics at HC III and above), adherence counselling	Danger signs, severe pneumonia, SAM, suspected TB in a child who cannot produce a sample, any child failing ART → HC IV
HC IV / General Hospital	POC EID and CD4, GeneXpert, admission for pneumonia and PCP, SAM care, TB-LAM, CrAg	Suspected meningitis without LP, failed first line needing a resistance test, no response to PCP treatment, needs L-AmB → RRH
RRH (Centres of Excellence)	LP, cryptococcal and TB meningitis care, chest X-ray, second-line ART, HDU	Third-line ART, complex resistance, malignancy (Kaposi, lymphoma), cardiomyopathy or nephropathy → NRH
NRH (Mulago), Baylor-Uganda, IDI	Specialist paediatric and adolescent HIV care, third line, oncology	n/a

Monitoring and follow-up

1. **Every visit**: weight → **re-check the dose band**; adherence (pill count, caregiver report), side-effects, TB screen, development, school, disclosure progress.
2. **Viral load**: at **6 months and 12 months** after starting ART, then **every 6 months** until 19 years.
3. **VL ≥1000** (or 201–999 on repeat): **intensive adherence counselling** (3 sessions, one a month, using the "3 Cs" good-score target >95%), address barriers, check doses and interactions, **repeat VL** after IAC (3 months).
4. **Still ≥1000 on a DTG or PI regimen → HIV drug resistance test**, then switch guided by the result.

5. **Failing an NNRTI regimen** → switch to DTG without waiting for a second VL (do IAC alongside).
6. **Second line (UCG 2023)**: failing **ABC + 3TC + DTG** → **AZT + 3TC + DRV/r** (alternative AZT + 3TC + LPV/r); failing **AZT + 3TC + DTG** → ABC + 3TC + DRV/r. Third line only via specialist centres with resistance testing.
7. **New positive PCR**: review in **2 weeks** after starting ART.

Disclosure, adherence and adolescents

Disclosure is planned with the caregiver and done in stages:

Stage	Age (guide)	What the child knows
Non-disclosure	Under about 5 years	"You take medicine to stay strong."
Partial disclosure	About **6–8 years** (school age)	"There are germs in your blood; the medicine keeps them asleep and makes your soldiers strong." The word "HIV" is not yet used
Full disclosure	About **10–12 years**, and **before adolescence/sexual debut**	Names HIV, how it is transmitted, that it is lifelong but controllable, who to tell and who not to
Post-disclosure support	Ongoing	Peer groups (OTZ, teen clubs), counselling, check understanding

Signs that a child is ready: asks why they take medicine, asks about blood tests, can keep a secret, cognitive maturity. **Delayed disclosure** is linked to poor adherence, depression and onward transmission in adolescence.

Adolescents (10–19 years) need:

- A **youth-friendly** clinic (separate day, peer supporters, flexible hours).
- **HEADSSS** psychosocial screening, **depression screening** (PHQ-9/PHQ-A), substance use, **sexual and reproductive health**: condoms, contraception (note EFV reduces implant efficacy; DTG has no such interaction), PrEP for partners, **HPV vaccination** (girls with HIV get **2 doses**, unlike the single dose for others).
- **Confidentiality** (see *Professionalism & Ethics*), **transition** to adult care at about 19–24 years, planned not abrupt.
- **Switch to TLD** at ≥30 kg; once-daily single tablet helps adherence.

10. Complications

Disease-related

- **Opportunistic infections**: PCP, TB, cryptococcal and TB meningitis, oesophageal candida, CMV, cryptosporidiosis.
- **Malnutrition and stunting**; delayed puberty.
- **Neurological**: HIV encephalopathy, developmental delay, cognitive and school problems.
- **Chronic lung disease**: LIP, bronchiectasis, post-TB lung disease.
- **Cardiac and renal**: cardiomyopathy, HIV-associated nephropathy.
- **Malignancy**: Kaposi sarcoma, non-Hodgkin and Burkitt lymphoma.
- **Psychosocial**: orphanhood, stigma, depression, suicide, school drop-out, abuse.

Treatment-related

Complication	Cause	Recognition and action
Hypersensitivity	Abacavir	Fever, rash, GI or respiratory symptoms in the first 6 weeks: stop, **never re-challenge**

Complication	Cause	Recognition and action
Anaemia, neutropenia	AZT (also CTX)	FBC; switch AZT
Rash, hepatitis	NVP	Stop; check liver function
CNS effects (dreams, dizziness)	EFV	Give at night; switch to DTG
Insomnia, headache, modest weight gain	DTG	Usually mild; give in the morning if insomnia
Renal toxicity, bone loss	TDF (adolescents)	Creatinine, urinalysis; avoid if CrCl <60
Diarrhoea, dyslipidaemia	LPV/r	Supportive; switch if intolerable
IRIS (unmasking TB, cryptococcus)	Immune recovery after starting ART	Deterioration 2–12 weeks after ART: treat the OI, **do not stop ART**
Treatment failure and drug resistance	Poor adherence, wrong dose, interactions	VL, IAC, resistance test, switch

11. Prognosis

- **Untreated**: about **30% die by 1 year and 50% by 2 years** (UCG 2023); about one in five infants are rapid progressors who die in the first year.
- **On early ART with suppression**: children grow, go to school and survive into adulthood; the risk shifts to adherence, adolescence and loss to follow-up.
- **Poor prognostic features**: late diagnosis, SAM, TB, cryptococcal meningitis, unsuppressed VL, non-disclosure and a disengaged adolescent.
- Long-term survivors may have stunting, chronic lung disease (LIP, bronchiectasis, post-TB) and cognitive problems.

12. Prevention

- **Primary (prevent infection)**: prevention of mother-to-child transmission (see *HIV in Pregnancy*): maternal testing and retesting, suppressed mothers, infant prophylaxis, exclusive breastfeeding (never mixed feeding under 6 months); prevention and recognition of sexual abuse; adolescent sexual health (condoms, PrEP for partners).
- **Secondary (find early)**: EID at 6 weeks, 9 months and 6 weeks after weaning; test every sick, malnourished or TB-affected child of unknown status; index-client testing of siblings ("Know Your Child's Status").
- **Tertiary (prevent the killers)**: same-day ART, co-trimoxazole, TB preventive therapy, immunisation, insecticide-treated nets, nutrition, viral load monitoring, disclosure and adolescent support.

13. Special Populations

- **Neonates (<4 weeks)**: AZT + 3TC + NVP (or RAL-based) until ≥4 weeks and ≥3 kg, then ABC + 3TC + DTG Verify *current neonatal option*.
- **Baby first seen after 6 weeks**: PCR plus first-line ART for 6 weeks (section 9).
- **TB co-infection**: TB treatment first, ART within 2 weeks; DTG twice daily on rifampicin; super-boost LPV/r; no NVP.
- **TB meningitis and cryptococcal meningitis**: defer ART (4–8 weeks and 4–6 weeks).
- **Severe acute malnutrition**: high mortality; treat SAM, start ART within 2 weeks once stable; screen for TB.

- **Epilepsy**: avoid carbamazepine, phenytoin and phenobarbital with DTG; use levetiracetam or valproate.
- **Adolescents**: TLD at ≥30 kg; HEADSSS, depression, contraception (EFV reduces implant efficacy), pregnancy, HPV 2 doses for girls with HIV, transition.
- **Immunisation in HIV**: no BCG if symptomatic HIV and missed at birth; avoid yellow fever in symptomatic HIV; give measles–rubella even if symptomatic.

14. Key Points

High-yield points

- Under 18 months, **DNA PCR** diagnoses HIV; a positive rapid test shows **exposure** only.
- EID **6 weeks → 9 months → 6 weeks after weaning → rapid test at 18 months**.
- Any positive PCR: **ART the same day**, confirmatory sample at the same time.
- **ABC + 3TC + DTG** from 4 weeks and 3 kg; DTG 50 mg from 20 kg; TLD from 30 kg; **dose by weight band, re-weigh every visit**.
- **DTG twice daily with rifampicin**, continued 2 weeks after rifampicin stops.
- **Co-trimoxazole** from 6 weeks for every HEI; for life in HIV-positive children.
- **PCP**: infant 2–6 months, hypoxic with a clear-ish chest: high-dose CTX for 21 days + prednisolone.
- **Defer ART** in cryptococcal and TB meningitis; otherwise never delay.
- VL at 6 and 12 months, then 6-monthly: ≤200 / 201–999 / ≥1000.
- **Every admitted child needs a known HIV status.**

Numbers & doses

Item	Value
Untreated infant mortality	**30% by 1 year, 50% by 2 years** (UCG 2023)
Diagnostic test under 18 months	**DNA PCR** (virological)
EID schedule	PCR **6 weeks**, **9 months**, **6 weeks after weaning**; rapid test **18 months**
NVP prophylaxis	**10 mg** (2.0–2.49 kg) / **15 mg** (≥2.5 kg) daily to 6 weeks; **20 mg** daily 6–12 weeks
High-risk infant	**AZT + NVP 6 weeks, then NVP 6 weeks** (12 weeks)
First line ≥4 weeks, ≥3 kg	**ABC + 3TC + DTG**
DTG 50 mg film-coated	From **20 kg**; TLD from **30 kg**
pDTG 10 mg dispersible	6–9.9 kg **1.5**; 10–13.9 kg **2**; 14–19.9 kg **2.5**; 20–24.9 kg **3** tabs; 3–5.9 kg ½ tab (FDA/WHO) or 1 (UCG table)
ABC/3TC 120/60	3–5.9 **1**; 6–9.9 **1.5**; 10–13.9 **2**; 14–19.9 **2.5**; 20–24.9 **3**; ≥25 kg 600/300 ×1
DTG + rifampicin	**Twice daily**, until 2 weeks after rifampicin ends
CTX prophylaxis (once daily)	<5 kg **120 mg**; 5–14.9 **240 mg**; 15–29.9 **480 mg**; ≥30 **960 mg**
Dapsone (CTX alternative)	**2 mg/kg** daily, max **100 mg**
PCP treatment	**TMP 8 mg/kg** (as CTX) **8-hourly for 21 days** + prednisolone **1–2 mg/kg/day** if hypoxic
INH preventive therapy	**10 mg/kg** (max 300 mg) daily for **6 months** + pyridoxine
VL schedule (0–19 y)	**6 and 12 months**, then **every 6 months**
Uganda VL thresholds	**≤200** suppressed; **201–999** LLV; **≥1000** non-suppressed
Virological failure	**2 VLs >1000**, 3–6 months apart, after IAC, ≥6 months on ART

Item	Value
Severe immunosuppression (CD4%)	<25% (<12 m); <20% (12–35 m); <15% (36–59 m); <200 cells (≥5 y)
AHD	**All children <5 years**, or CD4 <200, or stage 3/4
Disclosure	Partial **6–8 y**; full **10–12 y**

Exam traps

- **"Rapid test positive at 12 months = HIV-infected."** Wrong: it shows exposure; do a **DNA PCR**.
- **"PCR negative at 6 weeks, discharge."** Wrong if the baby is **breastfeeding**: the baby is still exposed.
- **"Wait for the confirmatory PCR before starting ART."** Wrong: start the **same day** and send the confirmatory sample at the same time.
- **"Start ART today" in cryptococcal meningitis.** Wrong: defer **4–6 weeks** (early ART raised mortality in trials).
- **"Swap pDTG 30 mg for DTG 50 mg mg-for-mg."** They are not bioequivalent; follow the table.
- **"Re-challenge abacavir after the rash settles."** Never: the re-challenge reaction can be fatal.
- **"Stop co-trimoxazole once the child is on ART."** In Uganda HIV-positive children continue CTX **for life** (UCG 2023).
- **"Oral thrush in a 2-week-old = stage 3."** Thrush counts only **after 6 weeks**.

Questions seniors ask

Viva questions

1. *Why can't you use a rapid test at 6 weeks?* Maternal IgG crosses the placenta and persists for up to 18 months; a positive result shows exposure only.
2. *What is the EID schedule?* PCR at 6 weeks, 9 months, 6 weeks after weaning; rapid test at 18 months.
3. *Preferred first-line for a 12 kg child?* ABC/3TC 120/60 two tablets + pDTG 10 mg two tablets (20 mg), once daily.
4. *When do you defer ART?* Cryptococcal meningitis (4–6 weeks) and TB meningitis (4–8 weeks); otherwise same day.
5. *How do you dose DTG with rifampicin?* Same band dose, twice daily, continued for 2 weeks after rifampicin stops.
6. *What are the components of HEI care?* Prophylaxis, EID, CTX, immunisation, feeding counselling, growth monitoring, TB screening, mother's care.
7. *What is LIP and how does it differ from miliary TB?* A lymphocytic infiltrate of the lungs in older HIV-positive children; clubbing, parotid swelling, well child, reticulonodular CXR; TB has fever, wasting and contact.

On the ward round

- **This 7-month-old's mother is HIV-positive and the baby had a negative PCR at 6 weeks. She is still breastfeeding. What now?** Continue CTX, check she completed prophylaxis, do the next PCR at 9 months and 6 weeks after breastfeeding stops, then a rapid test at 18 months; and check the mother's viral load.
- **The PCR is positive. What do you do today?** Counsel the caregiver, start ABC/3TC + pDTG by weight band, send a confirmatory DBS (plus an HIVDR sample if the mother is failing), screen for TB, give CTX, book review in 2 weeks.
- **Why do children fail ART more than adults?** They depend on a caregiver; the medicines taste bad; doses go out of date as they grow; there is non-disclosure, stigma at school and boarding; and earlier NNRTI exposure (resistance).

- **How would you manage a non-suppressed VL of 4,500 in a 10-year-old on ABC/3TC/DTG?** Three monthly IAC sessions exploring barriers (disclosure, school, caregiver), check the dose for weight and interactions, repeat VL; if still ≥1000, send a resistance test and switch per result (likely AZT + 3TC + DRV/r).
- **How do you recognise PCP clinically?** An infant aged 2–6 months, severe tachypnoea and hypoxaemia, little fever, a chest that sounds better than the saturation suggests, no response to first-line antibiotics; CXR shows bilateral ground glass.
- **When do you tell a child they have HIV?** Gradually: partial disclosure from about 6–8 years, full disclosure by 10–12 years, planned with the caregiver and done before adolescence, followed by peer support.
- **Which vaccines are different in a child with HIV?** No BCG if the child has symptomatic HIV and missed it at birth; avoid yellow fever in symptomatic HIV; give measles–rubella even if symptomatic; HPV is 2 doses for girls with HIV.

Mnemonics

- **EID "6–9–6–18"**: PCR at **6** weeks, **9** months, **6** weeks after weaning; rapid test at **18** months.
- **pDTG bands "½–1½–2–2½–3"** for 3–6–10–14–20 kg (the ½ for 3–5.9 kg follows the FDA/WHO label; UCG prints 1). **ABC/3TC "1–1½–2–2½–3"** for the same bands.
- **CTX "120–240–480–960"** for <5, 5–15, 15–30, >30 kg.
- **HEI care "PEACE-FM"**: **P**rophylaxis, **E**ID, **A**ntibiotic (CTX), **C**are of mother, **E**PI (immunise), **F**eeding counselling, **M**onitor growth.
- **Defer ART in "the two Ms"**: crypto **M**eningitis and TB **M**eningitis.
- **Failure checklist "A-D-I-R"**: **A**dherence, **D**ose (band out of date?), **I**nteractions, **R**esistance.

OSCE tips

- In a counselling station, **use the caregiver's words**, check understanding with **teach-back**, demonstrate dispersing pDTG, and always ask **"Who else in the family has been tested?"**
- In an examination station, plot growth as Z-scores, look for thrush, parotids, nodes and hepatosplenomegaly, and present with a WHO stage.

Presenting this tutorial

Edith presents: Week 2, Tuesday, 08:00–09:30, Dr Mawanda. Aim for **20–25 minutes** of talk and **10 minutes** of questions.

Suggested structure (about 12 slides or a whiteboard plan)

1. **Hook (1 min)**: a short case: "A 4-month-old HIV-exposed baby, SpO_2 82%, chest almost clear. What is killing him?" (PCP). Return to it at the end.
2. **Objectives (30 s)**: diagnose HIV in infants; start and dose ART; prevent OIs; keep children and adolescents in care.
3. **Burden and why children are different (2 min)**: vertical transmission routes; 30% die by 1 year untreated.
4. **Diagnosis (4 min)**: draw the **EID algorithm** on the board (6 weeks → 9 months → 6 weeks after weaning → 18 months). Explain why antibodies mislead. Include presumptive severe HIV disease.
5. **WHO staging (2 min)**: one example per stage; highlight the thrush-after-6-weeks and oesophageal-candida traps.
6. **ART (5 min)**: the regimen table by age/weight; the weight-band table for ABC/3TC and pDTG; DTG with rifampicin; ABC hypersensitivity.
7. **Prophylaxis (2 min)**: CTX bands; TPT; immunisation (no BCG in symptomatic HIV; MR even if symptomatic).
8. **OIs (3 min)**: PCP, TB, cryptococcal meningitis, thrush: recognition and doses.
9. **Monitoring and failure (2 min)**: VL schedule, Uganda thresholds (≤200 / 201–999 / ≥1000), IAC, switch.
10. **Disclosure and adolescents (2 min)**: stages of disclosure, OTZ, HEADSSS, transition.

11. **Back to the hook case (1 min)**: management of PCP.
12. **Take-home messages (30 s)**: five bullets (see High-yield points).

Delivery tips

- Bring a **real HEI card** and a **dosing chart** from the ART clinic if you can; consultants love local tools.
- Prepare answers for the questions consultants usually ask: *What if the mother refuses to test the baby? What if the PCR result never comes back? How do you dose DTG with rifampicin? What do you do when the child weighs 19.8 kg?* (Use the current band; re-weigh at every visit and move up when the next band is reached.)
- Say where Uganda differs from WHO (NVP-alone vs AZT + NVP for high-risk infants; suppression threshold 200 vs 50/1000), and that you checked the **UCG 2023** and **2022 consolidated guidelines**.
- End with **one MCQ to the audience**: it keeps the room awake.

15. References

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7. MSD Manual Professional Edition. Human Immunodeficiency Virus (HIV) Infection in Infants and Children. https://www.msdmanuals.com/professional/pediatrics/human-immunodeficiency-virus-hiv-infection-in-infants-and-children/human-immunodeficiency-virus-hiv-infection-in-infants-and-children
8. Glynn M, Drake WM, editors. Hutchison's Clinical Methods: An Integrated Approach to Clinical Practice. 24th edition. Edinburgh: Elsevier; 2018.

SELF-TEST

Flashcards

28 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Which test diagnoses HIV in a child under 18 months, and why not a rapid test?	**DNA PCR** (virological). Maternal IgG crosses the placenta and persists up to 18 months, so an antibody test shows only **exposure**.
2	Uganda EID schedule for an HIV-exposed breastfed infant	DNA PCR at **6 weeks** (or first contact), **9 months**, and **6 weeks after breastfeeding stops**; then a **rapid antibody test at 18 months**.
3	A DNA PCR comes back positive. What happens the same day?	Start **ART the same day** and send a **confirmatory sample** at the same time (plus an HIVDR sample if the mother is failing ART). Do not wait for the confirmatory result.
4	Untreated mortality of infants with HIV (UCG 2023)	About **30% die by 1 year** and **50% by 2 years**.
5	Preferred first-line ART for a child ≥4 weeks and ≥3 kg (Uganda 2022/UCG 2023)	**ABC + 3TC + DTG** (pDTG 10 mg dispersible or ABC/3TC/DTG 60/30/5 FDC). DTG 50 mg film-coated from **20 kg**; **TLD** from **30 kg**.
6	pDTG 10 mg dispersible doses by weight band	6–9.9 kg **1.5 tab** (15 mg); 10–13.9 kg **2** (20 mg); 14–19.9 kg **2.5** (25 mg); 20–24.9 kg **3** (30 mg) or DTG 50 mg ×1. 3–5.9 kg: FDA/WHO ½ tab (5 mg); UCG table prints 1 (verify with the MoH chart).
7	ABC/3TC 120/60 mg dispersible doses (once daily)	3–5.9 kg **1**; 6–9.9 kg **1.5**; 10–13.9 kg **2**; 14–19.9 kg **2.5**; 20–24.9 kg **3**; ≥25 kg ABC/3TC **600/300 one tablet**.
8	How is DTG dosed with rifampicin?	Same weight-band dose given **twice daily**, continued for **2 weeks after rifampicin stops**.
9	Are pDTG 10 mg dispersible and DTG 50 mg film-coated interchangeable mg for mg?	**No.** They are not bioequivalent: **30 mg pDTG ≈ 50 mg film-coated**.
10	Co-trimoxazole prophylaxis doses (once daily)	<5 kg **120 mg**; 5–14.9 kg **240 mg**; 15–29.9 kg **480 mg**; ≥30 kg **960 mg**.
11	When does CTX start and stop in an HIV-exposed infant?	Start at **6 weeks**; stop when HIV is **excluded after breastfeeding ends**. If infected: continue **for life** (UCG 2023).
12	Alternative if co-trimoxazole is contraindicated	**Dapsone 2 mg/kg once daily**, max **100 mg**.
13	Nevirapine prophylaxis doses (UCG 2023)	Birth–6 weeks: **10 mg** (2.0–2.49 kg) or **15 mg** (≥2.5 kg) once daily; 6–12 weeks: **20 mg** once daily.
14	Which infants are 'high-risk' HIV-exposed?	Mother on ART **≤4 weeks** before delivery, **VL >1000** in the 4 weeks before delivery, or diagnosed in the **3rd trimester or while breastfeeding**. Give AZT + NVP 6 weeks then NVP 6 weeks (12 weeks).
15	Viral load schedule for children and adolescents (0–19 years)	At **6 and 12 months** after starting ART, then **every 6 months**.
16	Uganda viral load thresholds	**≤200** copies/mL suppressed; **201–999** low-level viraemia; **≥1000** non-suppressed (IAC, repeat VL, resistance test).
17	Definition of virological failure	**Two consecutive VLs >1000 copies/mL**, 3–6 months apart, with intensive adherence support in between, after **≥6 months on ART**.
18	Classic clinical picture of PCP	Infant **2–6 months**, severe tachypnoea and **hypoxaemia**, little or no fever, chest **relatively clear**, no response to 48 h of antibiotics; CXR bilateral ground glass.

#	Question	Answer
19	PCP treatment (WHO Pocket Book 2013)	High-dose co-trimoxazole **TMP 8 mg/kg + SMX 40 mg/kg** per dose, IV or oral, **8-hourly for 21 days**, plus **prednisolone 1–2 mg/kg/day** for 1 week if hypoxic; then lifelong prophylaxis.
20	In which two OIs is ART deferred, and for how long?	**Cryptococcal meningitis** (4–6 weeks after antifungals) and **TB meningitis** (4–8 weeks). Other TB: start ART within 2 weeks.
21	WHO 2022 induction for cryptococcal meningitis	**Liposomal amphotericin B 10 mg/kg once**, plus **flucytosine 100 mg/kg/day** and **fluconazole 12 mg/kg/day** (max 800 mg) for **14 days**; then fluconazole consolidation 8 weeks.
22	Oral thrush: when is it WHO stage 3, and when stage 4?	Persistent **oral** candidiasis **after 6 weeks of age** = stage 3. **Oesophageal** candidiasis = stage 4.
23	Presumptive severe HIV disease in an antibody-positive infant <18 months	**Oral thrush + severe pneumonia** or **+ severe sepsis** (any two of these), or **any AIDS-defining condition**. Start ART and confirm with PCR.
24	Stages of disclosure to a child with HIV	Partial disclosure about **6–8 years** ('germs in the blood'), full disclosure about **10–12 years**, before adolescence; then ongoing support (OTZ, peer groups).
25	Abacavir hypersensitivity: what to do	Fever, rash, GI or respiratory symptoms, usually within 6 weeks: **stop ABC and never re-challenge** (re-challenge can kill).
26	Immunisation differences in HIV	No **BCG** if symptomatic HIV and missed at birth; avoid **yellow fever** if symptomatic; give **measles–rubella** even if symptomatic; **HPV 2 doses** for girls with HIV.
27	Who counts as having advanced HIV disease (WHO)?	**All children under 5 years**, plus anyone with **CD4 <200** or **WHO stage 3 or 4**.
28	Second-line ART for a child failing ABC + 3TC + DTG (UCG 2023)	**AZT + 3TC + DRV/r** (alternative AZT + 3TC + LPV/r), guided by a resistance test.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 10-month-old girl is brought to Mengo's young child clinic for a routine visit. Her mother has been on TLD since ANC; her last viral load 2 months ago was 'target not detected'. The baby had a negative DNA PCR at 6 weeks, completed NVP prophylaxis, takes co-trimoxazole and is still breastfeeding. She weighs 8.4 kg (WAZ −0.3) and is well. Today a rapid HIV antibody test was done by mistake and is POSITIVE. What is the MOST appropriate interpretation and action?

A She is HIV-infected; start ABC/3TC + pDTG today
B Repeat the rapid test with a second kit; if positive, start ART
C The result reflects maternal antibody; collect the 9-month DNA PCR now and continue co-trimoxazole
D Stop breastfeeding immediately and repeat the rapid test in 3 months
E She is HIV-negative because the 6-week PCR was negative; stop co-trimoxazole

2. A 3-year-old boy weighing 12.5 kg is diagnosed HIV-positive today at a HC IV by the national rapid-test algorithm; he stopped breastfeeding at 14 months. He has no TB symptoms and no signs of meningitis, and his grandmother is ready to start treatment today. Which regimen and dose is MOST appropriate?

A ABC/3TC 120/60 mg 2 tablets + pDTG 10 mg 2 tablets, both once daily
B TDF/3TC/DTG 300/300/50 mg one tablet once daily
C AZT/3TC 60/30 mg 2 tablets + NVP syrup 20 mg, twice daily
D ABC/3TC 120/60 mg 2.5 tablets + DTG 50 mg one tablet once daily
E ABC/3TC 120/60 mg 2 tablets + LPV/r 40/10 pellets 4 sachets once daily

3. A 4-month-old HIV-exposed boy (weight 5.2 kg) presents to Mengo with 4 days of cough and worsening breathing. He has been mixed-fed; his mother was diagnosed HIV-positive at delivery and has 'not started her drugs yet'. Temperature 37.6°C, RR 72/min, severe chest indrawing, SpO_2 81% in air, a few scattered crackles only. He has oral thrush. Besides oxygen, ampicillin and gentamicin, what is the MOST important additional treatment now?

A Nebulised salbutamol every 20 minutes
B Oral amoxicillin 40 mg/kg twice daily instead of the IV antibiotics
C Start ABC/3TC + pDTG today and wait for the antibiotics to work
D IV ceftriaxone and await the chest X-ray
E High-dose co-trimoxazole (TMP 8 mg/kg 8-hourly) plus prednisolone

4. A 9-year-old girl (24 kg) on ABC/3TC + DTG 50 mg is diagnosed with pulmonary TB (Xpert MTB detected, RIF not detected). She will start RHZE. What change to her ART is MOST appropriate?

A Stop ART until TB treatment is finished
B Give DTG 50 mg twice daily during rifampicin and for 2 weeks after
C Switch DTG to nevirapine for the duration of TB treatment
D Continue DTG 50 mg once daily; no change is needed
E Switch to TDF/3TC/DTG because TDF protects the liver

5. A 13-year-old boy (31 kg) on TLD for 2 years has a viral load of 4,800 copies/mL. Six months ago it was 'target not detected'. His aunt says he moved to a boarding school this term and 'doesn't want friends to see his tablets'. He has not been told his full diagnosis. What is the BEST next step?

A Switch immediately to AZT + 3TC + DRV/r

B Send an HIV drug-resistance test today and await the result before any counselling

C Begin intensive adherence counselling, plan full disclosure with the aunt, and repeat the viral load after 3 sessions

D Increase DTG to 50 mg twice daily

E Reassure the aunt; a single high viral load is a laboratory error

6. A 15-month-old child (7.1 kg, WHZ −3.2, no oedema) is admitted with SAM. The mother's HIV status is unknown and she is not present; the grandmother brought the child. What is the MOST appropriate HIV testing approach?

A Defer HIV testing until nutritional rehabilitation is complete

B Test the grandmother because she is the caregiver

C Do a DNA PCR only if the CD4 count is low

D Do a rapid antibody test on the child; if positive, send a DNA PCR to confirm infection

E Do a rapid antibody test on the child; if positive, start ART immediately without further tests

7. An HIV-positive 11-year-old girl (28 kg) started ABC/3TC + DTG 50 mg 3 weeks ago (CD4 at baseline 90 cells/µL). She now has fever, a new swollen tender cervical lymph node and worsening cough, though she says she takes every dose. Which is the MOST likely explanation? Advanced

A Abacavir hypersensitivity

B Virological failure due to DTG resistance

C Co-trimoxazole allergy

D Immune reconstitution inflammatory syndrome unmasking TB

E Lymphoid interstitial pneumonitis

8. A newborn (2.8 kg) is born at a HC IV to a mother who was diagnosed with HIV at 36 weeks and started TLD 2 weeks before delivery. She plans to breastfeed. What is the MOST appropriate infant prophylaxis?

A AZT + NVP once/twice daily for 6 weeks, then NVP alone for a further 6 weeks

B NVP 15 mg once daily for 6 weeks

C No prophylaxis because the mother is on TLD

D Co-trimoxazole from birth

E Full ART (ABC/3TC + pDTG) from birth

9. A 14-year-old girl on TLD presents to Mengo with 10 days of headache and 2 days of vomiting. She is drowsy (GCS 13), afebrile, with mild neck stiffness. Her CD4 last month was 70 cells/µL. Serum CrAg is positive. A lumbar puncture shows opening pressure 32 cm H_2O. Besides starting antifungals, what is the MOST important additional step? Advanced

A Give dexamethasone 0.15 mg/kg 6-hourly

B Stop TLD until the meningitis has fully resolved, then restart after 6 months

C Give mannitol and avoid further lumbar punctures

D Perform repeated therapeutic lumbar punctures to lower the pressure

E Start acetazolamide to reduce CSF production

10. A 7-year-old (21 kg) has been on ABC/3TC + DTG for 3 years and is suppressed. His mother asks when and how he should be told about his HIV. He has started asking why he 'takes medicine every day when other children don't'. What is the BEST advice?

A Wait until he is 18 so that he can consent as an adult

B Tell him today that he has HIV and explain sexual transmission

C Begin partial disclosure now ('germs in your blood that the medicine keeps asleep'), planning full disclosure by about 10–12 years

D Tell him the medicine is vitamins so that he does not tell others

E Leave disclosure to his teachers at school

11. A 5-year-old HIV-positive girl (16 kg) on ABC/3TC + pDTG is suppressed. She has had 3 years of slowly progressive cough. Examination: finger clubbing, bilateral painless parotid swelling, generalised lymphadenopathy, SpO_2 95%, no fever, good appetite. CXR shows bilateral reticulonodular shadowing with hilar fullness. Xpert on two induced sputa is negative. What is the MOST likely diagnosis? Advanced

A Miliary tuberculosis

B Lymphoid interstitial pneumonitis

C Pneumocystis pneumonia

D Kaposi sarcoma of the lung

E Staphylococcal pneumonia

12. An 18-month-old HIV-exposed boy (9.8 kg) who stopped breastfeeding at 12 months had negative DNA PCRs at 6 weeks and 9 months. What test is needed now to confirm his final status?

A A DNA PCR now

B A CD4 count

C A viral load

D No further test; he is negative already

E A rapid HIV antibody test

13. A 2-year-old HIV-positive boy (10.4 kg) started ABC/3TC + pDTG 5 weeks ago. He is brought back with fever 38.9°C, a generalised maculopapular rash, vomiting, abdominal pain and cough that began 2 days ago. What is the MOST appropriate action regarding his ART? Advanced

A Continue all ART and give paracetamol and an antihistamine

B Stop abacavir permanently and replace it with AZT, continuing 3TC and DTG

C Stop all ART for 2 weeks, then restart the same regimen

D Stop DTG because it commonly causes this reaction

E Stop abacavir now and restart it at half dose once the rash settles

14. A 6-week-old HIV-exposed baby (4.6 kg) attends for her first immunisations at a HC III. Her mother is on TLD and suppressed; the baby is exclusively breastfed and well. Which set of actions is MOST complete today?

A Give 6-week vaccines, take DBS for DNA PCR, start co-trimoxazole 120 mg once daily, continue NVP to 6 weeks as prescribed, and counsel on exclusive breastfeeding

B Give 6-week vaccines and ask the mother to return at 9 months for testing

C Take a rapid HIV test on the baby and start ART if positive

D Delay vaccines until the PCR result is back

E Start co-trimoxazole 240 mg once daily and advise stopping breastfeeding

15. Which statement about co-trimoxazole prophylaxis in Ugandan children is correct (UCG 2023)?

- A It is started at birth in all HIV-exposed infants
- B It is stopped once an HIV-positive child's viral load is suppressed
- C It is given at 120 mg once daily to a 12 kg child
- D It replaces the need for insecticide-treated nets
- E It continues for life in children who are HIV-infected

Answers & explanations

1. C · The result reflects maternal antibody; collect the 9-month DNA PCR now and continue co-trimoxazole

C is correct: under 18 months a positive antibody test shows exposure (maternal IgG), not infection. The 9-month PCR is due, and she stays on CTX while breastfeeding. A is wrong: starting ART on an antibody result in a well infant would wrongly label her as infected. B is wrong: a second antibody kit detects the same maternal IgG. D is wrong: Uganda recommends continued breastfeeding to 12 months while the mother is suppressed, and the next test is a PCR, not an antibody test. E is wrong: a negative 6-week PCR does not exclude infection in a breastfeeding baby who is still exposed.

2. A · ABC/3TC 120/60 mg 2 tablets + pDTG 10 mg 2 tablets, both once daily

A is correct: at 3 years he is diagnosed by rapid test (over 18 months and not breastfed for more than 6 weeks), and the 10–13.9 kg band takes ABC/3TC 120/60 ×2 + pDTG 10 mg ×2 (20 mg) once daily. B is wrong: TLD is for ≥30 kg; TDF is not used under 30 kg. C is wrong: NVP-based regimens are no longer preferred and fail more often. D is wrong: 2.5 tablets belongs to the 14–19.9 kg band and DTG 50 mg is used only from 20 kg. E is wrong: LPV/r is an alternative only if DTG is unavailable, and it is given twice daily, not once.

3. E · High-dose co-trimoxazole (TMP 8 mg/kg 8-hourly) plus prednisolone

E is correct: an HIV-exposed infant aged 2–6 months with severe hypoxaemia, little fever and a relatively clear chest is PCP until proven otherwise; WHO adds high-dose co-trimoxazole for 21 days and prednisolone 1–2 mg/kg/day if hypoxic. A is wrong: this is not bronchiolitis or asthma, and salbutamol will not treat PCP. B is wrong: severe pneumonia with SpO_2 81% needs parenteral therapy and admission. C is wrong: ART is important but the life-threatening PCP must be treated first; ART can start within 2 weeks once stable, and his PCR must be sent. D is wrong: ceftriaxone does not cover Pneumocystis, and waiting for imaging delays the key drug.

4. B · Give DTG 50 mg twice daily during rifampicin and for 2 weeks after

B is correct: rifampicin induces UGT1A1/CYP3A and lowers DTG levels, so the weight-band dose is given twice daily and continued for 2 weeks after rifampicin stops. A is wrong: interrupting ART causes rebound and resistance; ART continues. C is wrong: NVP must not be given with rifampicin. D is wrong: once-daily DTG with rifampicin gives sub-therapeutic levels. E is wrong: TDF is used from 30 kg and does not 'protect the liver'; it does nothing about the interaction.

5. C · Begin intensive adherence counselling, plan full disclosure with the aunt, and repeat the viral load after 3 sessions

C is correct: the first step for a non-suppressed VL (≥1000) is IAC (three monthly sessions) addressing the obvious barriers (boarding school, stigma, non-disclosure), then a repeat VL; only if it stays ≥1000 does a resistance test guide a switch. A is wrong: switching before IAC and a repeat VL on a DTG regimen is premature; resistance to DTG is uncommon. B is wrong: resistance testing comes after a second non-suppressed VL, and counselling should not wait. D is wrong: twice-daily DTG is for rifampicin co-treatment or known INSTI resistance, not adherence failure. E is wrong: ignoring a VL of 4,800 in an adolescent with clear adherence barriers risks resistance and disease progression.

6. D · Do a rapid antibody test on the child; if positive, send a DNA PCR to confirm infection

D is correct: when the mother is unavailable, UCG 2023 advises a rapid test on the child to establish exposure; under 18 months a positive result needs a DNA PCR to diagnose infection (although ART can start presumptively if criteria for severe HIV disease are met). A is wrong: HIV status changes SAM management and prognosis; test now. B is wrong: the grandmother's status says nothing about the child's exposure. C is wrong: CD4 is not a diagnostic test, and PCR does not depend on it. E is wrong: under 18 months a positive antibody test alone does not prove infection; confirm with PCR unless presumptive severe disease criteria are met.

7. D · Immune reconstitution inflammatory syndrome unmasking TB

D is correct: new inflammatory features (fever, enlarging node, cough) 2–12 weeks after starting ART in a child with a very low CD4 strongly suggest IRIS, usually unmasking TB; investigate for TB (Xpert, CXR, TB-LAM) and continue ART. A is wrong: abacavir hypersensitivity gives fever, rash and GI/respiratory symptoms, not a localised enlarging node; misdiagnosing it would mean stopping abacavir for ever without reason. B is wrong: virological failure cannot be diagnosed at 3 weeks and does not cause acute inflammation. C is wrong: sulfa allergy causes rash and fever, not lymphadenitis. E is wrong: LIP is a slow process with parotid swelling and clubbing, not an acute febrile illness after ART.

8. A · AZT + NVP once/twice daily for 6 weeks, then NVP alone for a further 6 weeks

A is correct: the mother has been on ART for under 4 weeks, so the baby is high-risk: AZT 15 mg twice daily + NVP 15 mg once daily for 6 weeks, then NVP for 6 more weeks (2022 consolidated guidelines; UCG 2023 lists NVP for 12 weeks). B is wrong: 6 weeks of NVP alone is for low-risk infants. C is wrong: maternal ART for only 2 weeks has not suppressed the virus. D is wrong: co-trimoxazole starts at 6 weeks and is not ARV prophylaxis. E is wrong: ABC/3TC + pDTG is not used in the first 4 weeks; full treatment is given to infants diagnosed as infected (or presenting after 6 weeks, per UCG).

9. D · Perform repeated therapeutic lumbar punctures to lower the pressure

D is correct: raised intracranial pressure is a major cause of death in cryptococcal meningitis; therapeutic LPs (remove CSF to halve pressure or below 20 cm H_2O) are repeated daily until pressure is controlled. A is wrong: steroids increase mortality in cryptococcal meningitis. B is wrong: this patient is already on ART; it is not stopped, and in a newly diagnosed patient ART is deferred only 4–6 weeks, not 6 months. C is wrong: mannitol does not help and LPs should be repeated, not avoided. E is wrong: acetazolamide was harmful in trials and is not recommended.

10. C · Begin partial disclosure now ('germs in your blood that the medicine keeps asleep'), planning full disclosure by about 10–12 years

C is correct: at school age a child who asks questions is ready for partial disclosure; full disclosure is planned with the caregiver by about 10–12 years, before adolescence. A is wrong: late disclosure is linked to poor adherence, depression and onward transmission; adolescents need to know before sexual debut. B is wrong: abrupt full disclosure including sexual transmission at 7, without preparation, is not age-appropriate. D is wrong: lying ('vitamins') destroys trust when he finds out and undermines adherence. E is wrong: disclosure belongs to the caregiver with the health team; telling teachers without consent breaches confidentiality.

11. B · Lymphoid interstitial pneumonitis

B is correct: LIP typically affects HIV-positive children over 2 years and causes a slow course, clubbing, parotid enlargement, lymphadenopathy, a well child and a reticulonodular CXR. A is wrong: miliary TB causes fever, wasting and an ill child, and Xpert or a contact history would usually point to it. C is wrong: PCP is an acute, hypoxic illness of infants aged 2–6 months with ground-glass changes. D is wrong: pulmonary KS is rare, usually with skin or palate lesions and effusions. E is wrong: staphylococcal pneumonia is acute and febrile with pneumatoceles or empyema.

12. E · A rapid HIV antibody test

E is correct: at 18 months, more than 6 weeks after weaning, maternal antibodies have gone and the Uganda algorithm ends with a rapid antibody test; if negative, he is discharged as HIV-negative and CTX is stopped. A is wrong: a PCR 6 weeks after weaning is part of the algorithm, but the final confirmation at 18 months is serological. B is wrong: CD4 does not diagnose HIV. C is wrong: viral load is a monitoring test, not the national diagnostic test for this purpose. D is wrong: the 9-month PCR was taken while he was still breastfeeding, so he needed further testing.

13. B · Stop abacavir permanently and replace it with AZT, continuing 3TC and DTG

B is correct: fever, rash and GI/respiratory symptoms within 6 weeks of starting abacavir suggest hypersensitivity; stop ABC permanently and substitute AZT (AZT + 3TC + DTG is the UCG alternative when ABC is contraindicated). A is wrong: continuing ABC through a hypersensitivity reaction can be fatal. C is wrong: restarting the same regimen re-challenges abacavir, which can cause a fatal reaction. D is wrong: DTG hypersensitivity is rare; the multi-system picture fits abacavir. E is wrong: abacavir must never be re-challenged at any dose.

14. A · Give 6-week vaccines, take DBS for DNA PCR, start co-trimoxazole 120 mg once daily, continue NVP to 6 weeks as prescribed, and counsel on exclusive breastfeeding

A is correct: at 6 weeks the HEI gets routine vaccines, the first DNA PCR, CTX at the <5 kg dose (120 mg), completion of NVP, and feeding counselling. B is wrong: the first PCR is due now, at 6 weeks. C is wrong: a rapid test at 6 weeks detects maternal antibody, so it cannot diagnose infection. D is wrong: vaccines should not be delayed for an HIV result; the baby is well. E is wrong: 240 mg is the 5–14.9 kg dose, and stopping breastfeeding is not advised when the mother is suppressed on ART.

15. E · It continues for life in children who are HIV-infected

E is correct: UCG 2023 states that infants who become HIV-infected continue co-trimoxazole for life. A is wrong: CTX starts at 6 weeks, not birth. B is wrong: in Uganda's guidance, viral suppression in a child is not a stopping criterion. C is wrong: a 12 kg child takes 240 mg (5–14.9 kg band); 120 mg is for under 5 kg. D is wrong: UCG advises both CTX and ITNs; together they reduce malaria risk by about 97%.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The hypoxic 4-month-old: PCP and presumptive HIV

At 22:00 you are called to the paediatric ward at Mengo Hospital. A 4-month-old boy, weight 4.9 kg, has had 5 days of cough and 2 days of fast breathing. Mother says she 'tested positive when he was born' but did not start ART; the baby got 'some syrup' for 2 weeks only and is mixed-fed. Temperature 37.4°C, RR 76/min, severe chest indrawing, SpO_2 80% in air, HR 168, capillary refill 2 s, RBS 4.2 mmol/L, alert.

1. What are your immediate priorities in the first 5 minutes?

> ETAT: he has an emergency sign (severe respiratory distress, SpO_2 80%). **Oxygen** by nasal prongs (0.5–1 L/min) or face mask to SpO_2 ≥94%, sit him up, gentle suction if needed. Check circulation (CRT 2 s, not shocked) and glucose (normal). IV access; send FBC, blood culture if available, and **DBS for DNA PCR**. Weigh him (4.9 kg) to dose drugs.

2. What is your differential diagnosis, and what is the most likely?

> Severe pneumonia in an HIV-exposed infant: (1) **PCP**: most likely (age 2–6 months, severe hypoxia, little fever, untreated mother, short prophylaxis, mixed feeding); (2) bacterial pneumonia (pneumococcus, Staph, Gram-negatives); (3) TB; (4) bronchiolitis/viral pneumonia; (5) heart failure. The disproportion between hypoxia and chest signs points to PCP.

3. Write the treatment with doses for this 4.9 kg infant.

> **Ampicillin 50 mg/kg (245 mg) IV 6-hourly + gentamicin 7.5 mg/kg (37 mg) IV once daily** (WHO severe pneumonia). **High-dose co-trimoxazole**: TMP 8 mg/kg ≈ **40 mg TMP / 200 mg SMX** IV or by NGT **8-hourly for 21 days**. **Prednisolone 2 mg/kg/day ≈ 10 mg** once daily for 1 week then taper (hypoxic). Oxygen, maintenance feeds (breast milk by NGT if distressed), careful fluids.

4. How do you establish his HIV status, and what does 'presumptive severe HIV disease' mean here?

> He is under 18 months, so a rapid test shows only exposure; send a **DNA PCR** (POC if available). If PCR is not available quickly and he is antibody-positive, WHO criteria for **presumptive severe HIV disease** are met by **oral thrush + severe pneumonia**, or any AIDS-defining illness such as PCP. That allows starting ART while awaiting PCR confirmation.

5. When and with what will you start ART?

> Once he is stable (within about 1–2 weeks of starting PCP treatment; there is no reason to wait as long as for meningitis): **ABC/3TC 120/60 mg 1 tablet + pDTG** for the 3–5.9 kg band (check the current MoH chart: FDA/WHO 5 mg = ½ tablet; UCG table prints 1 tablet), once daily, after counselling the mother. Send the confirmatory PCR at initiation.

6. What else must happen for this family before discharge?

> **Start the mother on TLD today** and check her VL in 3 months; screen her for TB and cryptococcus if unwell. Continue the baby on **CTX prophylaxis for life** after the 21-day course. Feeding counselling (exclusive breastfeeding to 6 months now that the mother is starting ART, no mixing). Catch-up immunisation (6- and 10-week doses). Test siblings and the partner (index testing). Link to the ART clinic and book review in 2 weeks with a dose check.

Take-away: In an HIV-exposed infant aged 2–6 months, severe hypoxaemia with a relatively clear chest is PCP: give high-dose co-trimoxazole and prednisolone alongside antibiotics, establish infection by PCR,

start ART within 2 weeks, and treat the mother.

Case 2: The non-suppressed adolescent at boarding school

A 14-year-old girl, weight 38 kg, attends the Mengo adolescent HIV clinic with her aunt. She acquired HIV perinatally, has been on TLD for 3 years, and her viral load is now 12,400 copies/mL (previously suppressed). She looks well. She started at a boarding secondary school in Wakiso this year.

1. How will you take her history? Who should be in the room?

> See her **with the aunt first** (adherence, school, regimen), then **alone** for a confidential **HEADSSS** assessment: Home, Education/employment, Activities, Drugs, Sexuality, Suicide/depression, Safety. Explain confidentiality and its limits (risk of serious harm, abuse).

2. She tells you she hides her tablets at school, often misses doses and has a boyfriend. What are the key issues?

> (1) **Non-adherence** from stigma and boarding (likely cause of non-suppression); (2) **possible incomplete disclosure/understanding**; (3) **sexual activity**: risk to partner (not suppressed), STIs, pregnancy; (4) possible **depression**; (5) consent and capacity for contraception and SRH services.

3. What is the management of her viral load result?

> **Intensive adherence counselling** (3 monthly sessions) with practical solutions: dose timing that fits school (e.g. evening dose), a **school nurse** or matron as supporter (with her consent), a pill box, phone reminders, OTZ/peer support, treat depression. Check for interacting drugs (antacids, iron, anticonvulsants). **Repeat VL after IAC**. If still ≥1000: **HIV drug-resistance test** and switch per result.

4. What sexual and reproductive health care does she need today?

> Counsel on **U=U** (undetectable = untransmittable) and that she is **not** currently suppressed: condoms every time. Offer **contraception** (DTG has no interaction with implants; EFV would). Screen for STIs and pregnancy. Offer **PrEP** for the partner via his own clinic. Check HPV vaccination (girls with HIV: 2 doses). Screen for **sexual abuse/coercion**: if the partner is an adult, this may be defilement under Ugandan law (she is under 18) and child-protection steps apply.

5. What is your plan for her long-term care?

> Enrol in an **OTZ club** and youth clinic day; **depression screening** (PHQ-A) and counselling; reinforce full disclosure and understanding; plan **transition** to adult care at about 19–24 years; VL every 6 months once suppressed; continue CTX per national guidance; annual review of school support.

6. The aunt asks you to tell her everything the girl said in private. How do you respond?

> Explain that adolescents are entitled to **confidentiality**, which helps them stay in care, and that you will share anything that puts her at **serious risk** (abuse, self-harm). Encourage the girl to share with her aunt and offer to help her do so. Share what the aunt needs to support adherence (the plan), not the private details.

Take-away: Most adolescent non-suppression is an adherence and psychosocial problem: see the teenager alone, use HEADSSS, give intensive adherence counselling with practical school solutions, address sexual health and safeguarding, and only then test for resistance.

Case 3: TB and HIV in a malnourished toddler

A 2-year-6-month-old boy is referred from a HC IV in Mpigi to Mengo. Weight 8.2 kg (WAZ −3.1), height 83 cm (HAZ −2.6), WHZ −2.5, MUAC 12.0 cm, no oedema. He has had cough and fever for 4 weeks and night sweats; his father is on TB treatment. A rapid HIV test today is positive; his mother died last year. Immunisation card: complete up to 9 months; missed MR2.

1. Is the positive rapid test enough to diagnose HIV here?

> **Yes.** He is over 18 months and no longer breastfeeding, so a positive antibody test (confirmed by the national algorithm with a second kit) diagnoses infection.

2. What investigations will you request and why?

> **GeneXpert** on gastric aspirate/induced sputum and **stool** (TB, RIF resistance); **CXR** (hilar nodes, miliary); **urine TB-LAM** (AHD: every child under 5 with HIV counts as AHD); **CD4%**; **FBC**; **malaria RDT**; RBS; **viral load** baseline is optional (it is used for monitoring).

3. Xpert on stool: MTB detected, RIF not detected. In what order do you start treatment?

> Start **TB treatment first** (weight-band RHZE per national/WHO 2022 guidance; a child with non-severe disease may qualify for the 4-month regimen), **co-trimoxazole 240 mg daily** (8.2 kg band), and nutritional support (RUTF for moderate wasting). **Start ART within 2 weeks** once TB drugs are tolerated, because he does not have TB meningitis.

4. Write his ART prescription including the TB interaction.

> 6–9.9 kg band: **ABC/3TC 120/60 mg 1.5 tablets once daily** + **pDTG 10 mg 1.5 tablets (15 mg) TWICE daily** while on rifampicin and for 2 weeks after it ends; then pDTG 1.5 tablets once daily. Re-weigh each visit and move to the 10–13.9 kg band (2 tabs each) when he reaches 10 kg.

5. Three weeks after starting ART his fever returns and his cervical nodes swell. He is taking all doses. What do you think and do?

> **TB-IRIS** (paradoxical): common 2–12 weeks after starting ART in TB co-infection. Exclude other causes (malaria, bacterial infection, TB drug resistance or poor TB adherence). **Continue ART and TB treatment**; give NSAIDs or **prednisolone 1–2 mg/kg/day** for severe IRIS; aspirate a fluctuant node.

6. What else does the family need?

> **Catch-up MR2** (measles–rubella can be given even with HIV); trace and test **siblings** and household (index testing, TB contact screening and TPT for contacts once TB is excluded); identify a stable **caregiver/treatment supporter** (orphan); link to OVC and social support; nutrition follow-up; disclosure planning later.

Take-away: In HIV–TB co-infection start TB treatment first and ART within 2 weeks (not in TB meningitis), double the DTG dose frequency while on rifampicin, expect IRIS, and remember the family: test contacts, nourish, immunise and find a caregiver.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.